Solutions

Maths

Q1. Consider two circles in the Cartesian plane defined by

$$C_1 : x^2 + y^2 = 25$$

$$C_2 : x^2 + y^2 - 10x + 21 = 0$$

Let L_1 and L_2 be the two direct common tangents to the circles C_1 and C_2, where L_1 touches C_1 and C_2 at the points P_1 and P_2 respectively, and L_2 touches C_1 and C_2 at the points Q_1 and Q_2 respectively. Let C_3 be the unique circle passing through the four points of tangency P_1, P_2, Q_1, and Q_2. If the line $L : 4x - 3y = 0$ intersects the circle C_3 at two distinct points A and B, then the length of the chord AB is equal to

(A) 6 (B) 7 (C) 8 (D) 9

Ans. B

Sol. The given circles are $C_1 : x^2 + y^2 = 25$ with center $O_1(0,0)$, radius $R_1 = 5$, and $C_2 : (x-5)^2 + y^2 = 4$ with center $O_2(5,0)$, radius $R_2 = 2$.

The external center of similitude T_e divides O_1O_2 externally in the ratio $5:2$: $T_e = \left(\frac{5(5) - 2(0)}{5-2}, 0\right) = \left(\frac{25}{3}, 0\right)$

Any line passing through T_e with slope m is given by $y = m\left(x - \frac{25}{3}\right) \implies 3mx - 3y - 25m = 0.$

Applying the condition of tangency to C_1: $\frac{|-25m|}{\sqrt{9m^2+9}} = 5 \Rightarrow 25m^2$

$= 9m^2 + 9 \Rightarrow m = \pm\frac{3}{4}$

Thus, the direct common tangents are $L_1 : 3x - 4y - 25 = 0$ and $L_2 : 3x + 4y - 25 = 0.$

The points of tangency P_1, Q_1 on C_1 are the feet of the perpendiculars from $(0,0)$ to L_1 and L_2: $P_1 = (3, -4), \quad Q_1 = (3, 4)$

Similarly, the points of tangency P_2, Q_2 on C_2 are computed from the center $(5,0)$: $P_2 = \left(\frac{31}{5}, -\frac{8}{5}\right), Q_2 = \left(\frac{31}{5}, \frac{8}{5}\right)$

Since P_1, P_2, Q_1, Q_2 are symmetric with respect to the x-axis, the center of circle C_3 lies on the x-axis at $(x_0, 0)$.

Equating the distances from $(x_0, 0)$ to $Q_1(3,4)$ and $Q_2\left(\frac{31}{5}, \frac{8}{5}\right)$:

$(3-x_0)^2 + 16 = \left(\frac{31}{5} - x_0\right)^2 + \frac{64}{25} \Rightarrow x_0 = \frac{5}{2}$

The radius squared is $R_3^2 = \left(3 - \frac{5}{2}\right)^2 + 16 = \frac{65}{4}$.

The equation of C_3 is: $\left(x - \frac{5}{2}\right)^2 + y^2 = \frac{65}{4}$

$\implies x^2 + y^2 - 5x - 10 = 0$

To find the points of intersection A and B with the line $L: 4x - 3y = 0 \implies y = \frac{4}{3}x$, substitute y into C_3:

$x^2 + \frac{16}{9}x^2 - 5x - 10 = 0$

$\Rightarrow 25x^2 - 45x - 90 = 0$

$\Rightarrow 5x^2 - 9x - 18 = 0$

$(5x + 6)(x - 3) = 0$

$\Rightarrow x_1 = 3, \quad x_2 = -\frac{6}{5}$

$AB = \sqrt{1 + m^2}|x_1 - x_2|$

$= \sqrt{1 + \frac{16}{9}}\left|3 - \left(-\frac{6}{5}\right)\right| = \frac{5}{3} \times \frac{21}{5} = 7$

Q2. Consider a function $g : [0, \infty) \to \mathbb{R}$ defined non-uniformly by

$$g(t) = \begin{cases} 4t - t^2, & 0 \le t \le 3 \\ t^2 - 8t + 18, & 3 < t \le 6 \\ 4t - 18, & t > 6 \end{cases}$$

Let the function $f : [0, \infty) \to \mathbb{R}$ be defined as $f(x) = \max_{t \in [0,x]} g(t)$ If S denotes the set of all points in the open interval $(0, \infty)$ at which $f(x)$ is NOT differentiable, then which of the following statements is **TRUE**?

(A) $S = \{2, 4 + \sqrt{2}, 6\}$ and $\int_0^2 f(x)dx = \frac{16}{3}$ (B) $S = \{4 + \sqrt{2}\}$ and $f'(2) + f'(6) = 4$

(C) $S = \{2, 4, 6\}$ and $\int_0^4 f(x)dx = \frac{40}{3}$ (D) $S = \{4 + \sqrt{2}, 6\}$ and $f'(2) = 0$

Ans. B

Sol. For $0 \le t \le 3$, $g'(t) = 4 - 2t$, which vanishes at $t = 2$. Thus, $g(t)$ strictly increases on $[0, 2]$ from $g(0) = 0$ to a local maximum $g(2) = 4$, and strictly decreases on $[2, 3]$ to $g(3) = 3$.
For $3 < t \le 6$, $g'(t) = 2t - 8$, revealing a local minimum at $t = 4$ with $g(4) = 2$.

For $t \in (4, 6]$, $g(t)$ strictly increases to $g(6) = 6$. For $t > 6$, $g'(t) = 4 > 0$, meaning $g(t)$ strictly increases for all $t > 6$.

$$f(x) = \begin{cases} 4x - x^2, & 0 \le x \le 2 \\ 4, & 2 < x < 4 + \sqrt{2} \\ x^2 - 8x + 18, & 4 + \sqrt{2} \le x \le 6 \\ 4x - 18, & x > 6 \end{cases}$$

* At $x = 2$: $\lim\limits_{x \to 2^-} f'(x) = 4 - 2(2) = 0$ and $\lim\limits_{x \to 2^+} f'(x) = 0$. Thus, $f(x)$ is differentiable at $x = 2$ with $f'(2) = 0$.

* At $x = 6$: $\lim\limits_{x \to 6^-} f'(x) = 2(6) - 8 = 4$ and $\lim\limits_{x \to 6^+} f'(x) = 4$. Thus, $f(x)$ is differentiable at $x = 6$ with $f'(6) = 4$.

* At $x = 4 + \sqrt{2}$: The left derivative is 0, while the right derivative is $2(4 + \sqrt{2}) - 8 = 2\sqrt{2}$. $f(x)$ is non-differentiable at this point.

Hence, the set of non-differentiability points is uniquely $S = \{4 + \sqrt{2}\}$, and $f'(2) + f'(6) = 0 + 4 = 4$.

Q3. Consider two fixed circles $S_1 : x^2 + y^2 - 1 = 0$ and $S_2 : x^2 + y^2 - 6x + 8 = 0$. Let S_3 be a variable circle passing through the point $(3, 0)$ and touching the line $x + 3 = 0$. Let P be the radical center of the circles S_1, S_2, and S_3. If the power of the point P with respect to the circle S_1 is minimum, then the length of the tangent drawn from P to the circle S_3 is equal to

(A) $\dfrac{\sqrt{5}}{2}$ (B) $\dfrac{5}{4}$ (C) $\sqrt{5}$ (D) $\dfrac{5}{2}$

Ans. A

Sol. The radical center $P(x_0, y_0)$ must lie on the radical axis of S_1 and S_2.

$S_1 - S_2 = 0 \implies 6x - 9 = 0 \implies x_0 = \dfrac{3}{2}$

Let the center of the variable circle S_3 be (h, k).

Since S_3 passes through $(3, 0)$ and touches the line $x = -3$, the locus of its center is a parabola with focus $(3, 0)$ and directrix $x = -3$.

$(h - 3)^2 + k^2 = (h + 3)^2 \implies k^2 = 12h$

The radius of S_3 is $h + 3$, so its equation is: $(x - h)^2 + (y - k)^2 = (h + 3)^2$

$\Rightarrow x^2 + y^2 - 2hx - 2ky + 6h - 9 = 0$

The radical axis of S_1 and S_3 is given by: $S_1 - S_3 = 0 \Rightarrow hx + ky - 3h + 4 = 0$

Since $P(x_0, y_0)$ lies on this line, substituting $x_0 = \dfrac{3}{2}$ and $h = \dfrac{k^2}{12}$ gives:

$\dfrac{k^2}{12}\left(\dfrac{3}{2}\right) + ky_0 - 3\left(\dfrac{k^2}{12}\right) + 4 = 0$

$\Rightarrow ky_0 - \dfrac{k^2}{8} + 4 = 0$

For $k \neq 0$, the y-coordinate of the radical center is: $y_0 = \dfrac{k}{8} - \dfrac{4}{k}$

The power of the radical center $P\left(\frac{3}{2}, y_0\right)$ with respect to S_1 is:

$\text{Power} = x_0^2 + y_0^2 - 1 = \frac{9}{4} + y_0^2 - 1 = y_0^2 + \frac{5}{4}$

To minimize the power, we must have $y_0 = 0$, which yields $k^2 = 32$ (admitting real values for k).

Thus, the minimum power of the radical center with respect to S_1 (and consequently with respect to S_3) is $\frac{5}{4}$.

The length of the tangent from P to S_3 is: $L = \sqrt{\text{Power}} = \sqrt{\frac{5}{4}} = \frac{\sqrt{5}}{2}$

Q4. Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous periodic function with period π, defined by $f(x) = \frac{1}{1+\sin^2 x}$ for all $x \in [0, \pi]$. Consider the definite integral $I = \int_0^{2026\pi} \frac{x^2 f(x)}{1+x^2} dx$ Which of the following inequalities is satisfied by I?

(A) $1013\sqrt{2}\pi - \tan^{-1}(2026\pi) < I < 1013\sqrt{2}\pi - \frac{1}{2}\tan^{-1}(2026\pi)$

(B) $1013\sqrt{2}\pi - \frac{1}{2}\tan^{-1}(2026\pi) < I < 1013\sqrt{2}\pi$

(C) $2026\sqrt{2}\pi - \tan^{-1}(2026\pi) < I < 2026\sqrt{2}\pi - \frac{1}{2}\tan^{-1}(2026\pi)$

(D) $2026\sqrt{2}\pi - \frac{1}{2}\tan^{-1}(2026\pi) < I < 2026\sqrt{2}\pi$

Ans. A

Sol. $\frac{x^2 f(x)}{1+x^2} = \frac{(1+x^2-1)f(x)}{1+x^2} = f(x) - \frac{f(x)}{1+x^2}$

$\int_0^{2026\pi} f(x)dx = 2026 \int_0^{\pi} \frac{1}{1+\sin^2 x} dx$

$\int_0^{\pi} \frac{1}{1+\sin^2 x} dx = 2\int_0^{\pi/2} \frac{\sec^2 x}{\sec^2 x + \tan^2 x} dx$

$= 2\int_0^{\pi/2} \frac{\sec^2 x}{1+2\tan^2 x} dx$

$2\int_0^{\infty} \frac{1}{1+u^2} \frac{du}{\sqrt{2}} = \sqrt{2}\left[\tan^{-1} u\right]_0^{\infty}$

$= \sqrt{2}\left(\frac{\pi}{2}\right) = \frac{\pi}{\sqrt{2}}$

$\int_0^{2026\pi} f(x)dx = 2026 \cdot \frac{\pi}{\sqrt{2}} = 1013\sqrt{2}\pi$

For the remaining part, let $J = \int_0^{2026\pi} \frac{f(x)}{1+x^2} dx$.

Since $0 \le \sin^2 x \le 1 \implies \frac{1}{2} \le f(x) \le 1$

$\frac{1}{2(1+x^2)} \le \frac{f(x)}{1+x^2} \le \frac{1}{1+x^2}$

$$\frac{1}{2}\tan^{-1}(2026\pi) < J < \tan^{-1}(2026\pi)$$

Recombining these sections under $I = 1013\sqrt{2}\pi - J$ reverses the inequality bounds:

$$1013\sqrt{2}\pi - \tan^{-1}(2026\pi) < I < 1013\sqrt{2}\pi - \frac{1}{2}\tan^{-1}(2026\pi)$$

Q5. Consider the circle $C : x^2 + y^2 = 4$ and the straight line $L : 3x + 4y = 25$. Let P be a variable point lying on the line L. From P, a pair of tangents are drawn to the circle C, touching it at the points A and B. Then which of the following statement(s) is (are) **TRUE**?

(A) The chord of contact AB always passes through the fixed point $\left(\frac{12}{25}, \frac{16}{25}\right)$.

(B) The locus of the circumcentre of $\triangle PAB$ is the line $6x + 8y = 25$.

(C) The locus of the midpoint of the chord AB is a circle passing through the origin.

(D) The minimum length of the chord AB occurs when OP is perpendicular to the line L (where O is the origin).

Ans. A, B, C, D

Sol. The equation of the chord of contact AB from $P(x_1, y_1)$ to the circle C is given by $T = 0$:
$xx_1 + yy_1 = 4$

Substituting $y_1 = \frac{25 - 3x_1}{4}$ into the equation: $xx_1 + y\left(\frac{25 - 3x_1}{4}\right) = 4$

$\Rightarrow x_1(4x - 3y) + (25y - 16) = 0$

This represents a family of lines passing through the intersection of $4x - 3y = 0$ and $25y - 16 = 0$.

Thus, the chord of contact always passes through the fixed point $\left(\frac{12}{25}, \frac{16}{25}\right)$.

Since $\angle OAP = \angle OBP = 90^\circ$, the quadrilateral $OAPB$ is cyclic, and the circumcircle of $\triangle PAB$ has OP as its diameter.
Let the circumcentre be $M(h, k)$, which is the midpoint of OP:
$h = \frac{x_1}{2}, \quad k = \frac{y_1}{2} \implies x_1 = 2h, \quad y_1 = 2k$

Substituting x_1 and y_1 into $3x_1 + 4y_1 = 25$:

$3(2h) + 4(2k) = 25 \implies 6h + 8k = 25$

Replacing (h, k) with (x, y), the locus is the straight line $6x + 8y = 25$. **Statement (B) is correct.**

Let $N(x_0, y_0)$ be the midpoint of the chord AB.

The equation of the chord in terms of its midpoint is $T = S_1$: $xx_0 + yy_0 = x_0^2 + y_0^2$

Comparing this with the chord of contact equation $xx_1 + yy_1 = 4$, we get:
$\frac{x_0}{x_1} = \frac{y_0}{y_1} = \frac{x_0^2 + y_0^2}{4}$

$\Rightarrow x_1 = \frac{4x_0}{x_0^2 + y_0^2}, \quad y_1 = \frac{4y_0}{x_0^2 + y_0^2}$

Substituting these into $3x_1 + 4y_1 = 25$: $3\left(\frac{4x_0}{x_0^2+y_0^2}\right) + 4\left(\frac{4y_0}{x_0^2+y_0^2}\right) = 25$

$\Rightarrow 25(x_0^2 + y_0^2) - 12x_0 - 16y_0 = 0$

The locus is the circle $25(x^2 + y^2) - 12x - 16y = 0$, which clearly passes through the origin $(0, 0)$.

Let $d = OP$ be the distance from the origin to the point P.

The length of the chord of contact AB is given by: $AB = \frac{2R\sqrt{d^2 - R^2}}{d}$

$= \frac{2(2)\sqrt{d^2 - 4}}{d} = 4\sqrt{1 - \frac{4}{d^2}}$

To minimize the length of AB, the value of $1 - \frac{4}{d^2}$ must be minimized, which requires minimizing d.

The minimum distance d from the origin to the line L occurs when OP is perpendicular to L.

Q6. Consider the function $f : \mathbb{R} \to \mathbb{R}$ defined by:

$$f(x) = \lim_{n\to\infty} \frac{x^{2n}(x^4 - 2x^2 + 2) + (\cos(\pi x) + 2)}{x^{2n} + 1}$$

Which of the following statements is (are) **TRUE**?

(A) The function $f(x)$ is differentiable on $\mathbb{R}$, but its second derivative $f''(x)$ does not exist at $x = 1$ and $x = -1$.

(B) $\lim\limits_{x\to 0} \frac{f(x) - 3 + \frac{\pi^2 x^2}{2}}{x^4} = \frac{\pi^4}{24}$.

(C) According to Lagrange's Mean Value Theorem applied to $f(x)$ on the interval $[0, 2]$, there exists a unique $c \in (0, 2)$ such that $f'(c) = \frac{7}{2}$, and this c lies in the subinterval $(1, 2)$.

(D) The number of points $c \in (-2, 2)$ satisfying $f'(c) = 0$ is exactly 3.

Ans. A, B, C, D

Sol. $f(x) = \begin{cases} \cos(\pi x) + 2 & \text{for } |x| < 1 \\ x^4 - 2x^2 + 2 & \text{for } |x| > 1 \\ 1 & \text{for } x = \pm 1 \end{cases}$

Continuity at $x = 1$: $\lim\limits_{x\to 1^-} f(x) = \cos(\pi) + 2 = 1$, and $\lim\limits_{x\to 1^+} f(x) = 1 - 2 + 2 = 1$.

Since $f(1) = 1$, f is continuous.

Differentiability at $x = 1$: $f'_-(1) = \lim\limits_{h\to 0^+} \frac{\cos(\pi(1-h)) + 2 - 1}{-h}$

$= \lim\limits_{h\to 0^+} \frac{1 - \cos(\pi h)}{-h} = 0$

$f'_+(1) = \lim\limits_{h\to 0^+} \frac{(1+h)^4 - 2(1+h)^2 + 2 - 1}{h}$

$= \lim\limits_{h\to 0^+} \frac{((1+h)^2 - 1)^2}{h} = 0$

Thus, $f'(1) = 0$. By symmetry, $f'(-1) = 0$. Hence, $f(x)$ is differentiable on $\mathbb{R}$.

$f''_-(1) = \lim\limits_{h\to 0^+} \dfrac{-\pi\sin(\pi(1-h)) - 0}{-h} = \pi^2$ & $f''_+(1) = \lim\limits_{h\to 0^+} \dfrac{4(1+h)^3 - 4(1+h) - 0}{h} = 8$

Since $\pi^2 \neq 8$, $f''(1)$ does not exist. By symmetry, $f''(-1)$ does not exist. Statement (A) is true.

For $x \to 0$, $|x| < 1$, so $f(x) = \cos(\pi x) + 2$.

$\lim\limits_{x\to 0} \dfrac{\cos(\pi x) - 1 + \frac{\pi^2 x^2}{2}}{x^4} = \lim\limits_{x\to 0} \dfrac{\left(1 - \frac{\pi^2x^2}{2} + \frac{\pi^4x^4}{24} - \ldots\right) - 1 + \frac{\pi^2x^2}{2}}{x^4} = \dfrac{\pi^4}{24}$; Statement (B) is true.

As $f(x)$ is continuous on $[0,2]$ and differentiable on $(0,2)$, Lagrange's Mean Value Theorem guarantees at least one $c \in (0,2)$ such that:

$f'(c) = \dfrac{f(2) - f(0)}{2 - 0} = \dfrac{10 - 3}{2} = \dfrac{7}{2}$

Setting $4c^3 - 4c = \dfrac{7}{2} \implies 8c^3 - 8c - 7 = 0$. Let $h(x) = 8x^3 - 8x - 7$.

We have $h(1) = -7 < 0$ and $h(2) = 41 > 0$.

Since $h'(x) = 24x^2 - 8 > 0$ for $x > 1$, $h(x)$ is strictly increasing.

Thus, a unique root exists in $(1,2)$. Statement (C) is true.

For Statement (D):

We solve $f'(x) = 0$ for $x \in (-2,2)$:

For $x \in (-2,-1)$, $4x(x^2-1) < 0$. No solution. At $x = -1$, $f'(-1) = 0$.

For $x \in (-1,1)$, $-\pi\sin(\pi x) = 0 \implies x = 0$. At $x = 1$, $f'(1) = 0$.

For $x \in (1,2)$, $4x(x^2-1) > 0$. No solution.

The solutions are exactly $c \in \{-1, 0, 1\}$, making exactly 3 points. Statement (D) is true.

Q7. Consider the function $f : \mathbb{R} \to \mathbb{R}$ defined by

$$f(x) = \int_{-x}^{x} \frac{t^2 + \cos t}{1 + e^t} dt$$

Which of the following statements is/are **CORRECT**?

(A) $f(x)$ is a strictly increasing function on $\mathbb{R}$. (B) $f(x)$ is an odd function.

(C) $f(\pi) = \dfrac{\pi^3}{3}$. (D) $\lim\limits_{x\to 0} \dfrac{f(x)}{x} = 0$.

Ans. A, B, C

Sol. $f'(x) = \dfrac{x^2 + \cos x}{1 + e^x} \cdot (1) - \dfrac{(-x)^2 + \cos(-x)}{1 + e^{-x}} \cdot (-1)$

$f'(x) = \dfrac{x^2 + \cos x}{1 + e^x} + \dfrac{x^2 + \cos x}{1 + e^{-x}}$

$= (x^2 + \cos x)\left(\dfrac{1}{1 + e^x} + \dfrac{e^x}{e^x + 1}\right)$

$= x^2 + \cos x$

Since $x^2 + \cos x > 0$ for all $x \in \mathbb{R}$, $f(x)$ is strictly increasing on $\mathbb{R}$.

Integrating $f'(x)$ gives: $f(x) = \dfrac{x^3}{3} + \sin x + C$

Since $f(0) = 0$, we have $C = 0$. Thus, $f(x) = \dfrac{x^3}{3} + \sin x$.

Clearly, $f(-x) = -f(x)$, so $f(x)$ is an odd function.

Evaluating at $x = \pi$: $f(\pi) = \dfrac{\pi^3}{3} + \sin \pi = \dfrac{\pi^3}{3}$

Evaluating the limit: $\lim\limits_{x \to 0} \dfrac{f(x)}{x} = \lim\limits_{x \to 0} \left(\dfrac{x^2}{3} + \dfrac{\sin x}{x} \right) = 1 \neq 0$

Q8. Let $f : \mathbb{R} \to \mathbb{R}$ be a function defined by

$$f(x) = \begin{vmatrix} x & 1 & 1 \\ -1 & x & 1 \\ -1 & -1 & x \end{vmatrix}$$

Let $g : \mathbb{R} \to \mathbb{R}$ be the inverse function of $f(x)$. Consider the function $D(x)$ defined by the determinant

$$D(x) = \begin{vmatrix} g(x) & 1 & 0 \\ g'(x) & g(x) & 1 \\ g''(x) & g'(x) & g(x) \end{vmatrix}$$

Which of the following statements is (are) **TRUE** ?

(A) $g'(4) = \dfrac{1}{6}$ (B) $g''(4) = -\dfrac{1}{72}$ (C) $D(4) = \dfrac{23}{36}$ (D) $D'(4) = \dfrac{55}{108}$

Ans. A, C, D

Sol. $f(x) = x(x^2 + 1) - 1(-x + 1) + 1(1 + x) = x^3 + 3x$

Since $f'(x) = 3x^2 + 3 > 0$ for all $x \in \mathbb{R}$, $f(x)$ is strictly increasing and thus bijective, ensuring that the inverse function $g(x)$ uniquely exists.

To find the values at $y = 4$, we set $f(x) = 4 \implies x^3 + 3x = 4$, which yields the unique real root $x = 1$.

Hence, $g(4) = 1$.

Differentiating $f(x)$ successively gives: $f'(x) = 3x^2 + 3 \Rightarrow f'(1) = 6$;
$f''(x) = 6x \Rightarrow f''(1) = 6$; $f'''(x) = 6 \Rightarrow f'''(1) = 6$

$g'(f(x)) = \dfrac{1}{f'(x)}$

$\Rightarrow g'(4) = \dfrac{1}{f'(1)} = \dfrac{1}{6}$

Differentiating $g'(f(x))f'(x) = 1$ with respect to x: $g''(f(x))(f'(x))^2 + g'(f(x))f''(x) = 0$

$\Rightarrow g''(f(x)) = -\dfrac{f''(x)}{(f'(x))^3}$

Evaluating at $x = 1$ gives: $g''(4) = -\dfrac{6}{6^3} = -\dfrac{1}{36}$

Expanding the determinant $D(x)$ along the first row:
$D(x) = g(x)(g(x)^2 - g'(x)) - 1(g'(x)g(x) - g''(x))$

$= g(x)^3 - 2g(x)g'(x) + g''(x)$

$D(4) = 1^3 - 2(1)\left(\frac{1}{6}\right) + \left(-\frac{1}{36}\right)$

$= 1 - \frac{1}{3} - \frac{1}{36} = \frac{23}{36}$

Differentiating $g''(f(x)) = -\frac{f''(x)}{(f'(x))^3}$ with respect to x:

$g'''(f(x))f'(x) = \frac{-f'''(x)(f'(x))^3 + 3(f''(x))^2(f'(x))^2}{(f'(x))^6}$

$= \frac{3(f''(x))^2 - f'''(x)f'(x)}{(f'(x))^4}$

$\Rightarrow g'''(f(x)) = \frac{3(f''(x))^2 - f'''(x)f'(x)}{(f'(x))^5}$

$g'''(4) = \frac{3(6^2) - 6(6)}{6^5} = \frac{2}{6^3} = \frac{1}{108}$

$D'(x) = 3g(x)^2g'(x) - 2g'(x)g''(x) - 2(g'(x)g''(x) + g(x)g''(x)) + g'''(x)$

$D'(x) = 3g(x)^2g'(x) - 2(g'(x))^2 - 2g(x)g''(x) + g'''(x)$

$D'(4) = 3(1)^2\left(\frac{1}{6}\right) - 2\left(\frac{1}{6}\right)^2 - 2(1)\left(-\frac{1}{36}\right) + \frac{1}{108} = \frac{1}{2} - \frac{2}{36} + \frac{2}{36} + \frac{1}{108} = \frac{55}{108}$

Paragraph

Consider a curve C defined parametrically for t > 0 by the equations:

$x(t) = \ln(t^2 + 1) - \frac{t^2}{t^2 + 1}$

$y(t) = \frac{t^4}{4} - t^2 + \frac{3}{2}\ln(t^2 + 1) + \frac{1}{2(t^2 + 1)}$

Let f(x) denote the slope of the tangent to the curve C at any point (x, y).

Q9. The value of $\frac{df}{dx}$ at the point on the curve where the tangent is parallel to the line 2y - x = 5 is

(A) 2 (B) 4 (C) 8 (D) 1

Ans. B

Sol. $\frac{dx}{dt} = \frac{2t}{t^2 + 1} - \frac{2t(t^2 + 1) - 2t^3}{(t^2 + 1)^2} = \frac{2t^3}{(t^2 + 1)^2}; \frac{dy}{dt} = t^3 - 2t + \frac{3t}{t^2 + 1} - \frac{t}{(t^2 + 1)^2} = \frac{t^7}{(t^2 + 1)^2}$

$f(x) = \frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{t^7}{2t^3} = \frac{t^4}{2}$

$\frac{df}{dx} = \frac{df/dt}{dx/dt} = \frac{2t^3}{\frac{2t^3}{(t^2+1)^2}} = (t^2 + 1)^2$

The tangent is parallel to 2y – x = 5, so slope = $\frac{1}{2}$: $\frac{t^4}{2} = \frac{1}{2} \Rightarrow t = 1 \quad (\text{since} t > 0)$

$\frac{df}{dx}|_{t=1} = (1^2 + 1)^2 = 4$

Q10. If $g(x) = f(x) - (e^2)x$, then the interval in which $g(x)$ is strictly increasing is

(A) $\left(0, \frac{1}{e}\right)$ (B) $(0, e)$ (C) $\left(\frac{1}{e}, \infty\right)$ (D) (e, ∞)

Ans. C

Sol. $\frac{dg}{dx} = \frac{df}{dx} - e^2 = (t^2+1)^2 - e^2$

For g(x) to be strictly increasing, we require $\frac{dg}{dx} > 0 : (t^2+1)^2 - e^2 > 0$

$\Rightarrow t^2 + 1 > e \Rightarrow t > \sqrt{e-1}$

Since $\frac{dx}{dt} = \frac{2t^3}{(t^2+1)^2} > 0$ for all t > 0, x(t) is a strictly increasing function.

Thus, the condition $t > \sqrt{e-1}$ corresponds to $x > x(\sqrt{e-1})$.

Evaluating x(t) at $t = \sqrt{e-1} : x(\sqrt{e-1}) = \ln(e) - \frac{e-1}{e}$

$= 1 - 1 + \frac{1}{e} = \frac{1}{e}$

Hence, g(x) is strictly increasing for $x \in \left(\frac{1}{e}, \infty\right)$.

Q1. Consider a real-valued function $y = f(x)$ defined implicitly by the equation $y^2 + \sin(xy) + x = y$ in a neighborhood of $x = 0$ such that $f(0) = 0$. If $f''(0)$ denotes the second-order derivative of $f(x)$ at $x = 0$, then the value of $f''(0)$ is _____.

Ans. 4

Sol. $y^2 + \sin(xy) + x = y$

Differentiating both sides with respect to x,

$2yy' + \cos(xy)(y + xy') + 1 = y'$

Substituting $x = 0$ and $y = 0$ into the equation: $2(0)y'(0) + \cos(0)(0 + 0) + 1 = y'(0)$

$\Rightarrow y'(0) = 1$

Differentiating the first derivative relation once more with respect to x:

$2(y')^2 + 2yy'' - \sin(xy)(y + xy')^2 + \cos(xy)(2y' + xy'') = y''$

Substituting the initial values $x = 0$, $y = 0$, and $y' = 1$: $2(1)^2 + 2(0)y''(0) - \sin(0)$

$(0 + 0)^2 + \cos(0)(2(1) + 0) = y''(0)$

$2 + 2 = y''(0) \implies y''(0) = 4$

Q2. Let $[t]$ and $\{t\}$ denote the greatest integer function and fractional part function of t, respectively. For $x \in (0,1) \setminus \left\{\frac{1}{n} : n \in \mathbb{N}\right\}$, consider the real-valued functions:

$\Phi(x) = \frac{1 - x^2\left[\frac{1}{x}\right]^2 - x^2\left\{\frac{1}{x}\right\}^2}{2x\left[\frac{1}{x}\right]}$ and $\Psi(x) = \frac{\left\{\frac{1}{x}\right\}}{\left[\frac{1}{x}\right] + \left\{\frac{1}{x}\right\}}$

If $L = \lim\limits_{x \to 0^+} \left(\frac{2^{\Phi(x)} + 8^{\Phi(x)} + 32^{\Phi(x)}}{3}\right)^{\frac{1}{\Psi(x)}}$ then the value of L is _____.

Ans. 8

Sol. Let $y = \dfrac{1}{x}$. Since $x \notin \left\{\dfrac{1}{n} : n \in \mathbb{N}\right\}$, $y \notin \mathbb{N}$.

Using $y = [y] + \{y\}$, we have $y^2 = [y]^2 + \{y\}^2 + 2[y]\{y\}$, which gives:

$y^2 - [y]^2 - \{y\}^2 = 2[y]\{y\}$

$\Phi(x) = \dfrac{x^2\left(\frac{1}{x^2} - \left[\frac{1}{x}\right]^2 - \left\{\frac{1}{x}\right\}^2\right)}{2x\left[\frac{1}{x}\right]}$

$= \dfrac{x^2\left(y^2 - [y]^2 - \{y\}^2\right)}{2x[y]} = \dfrac{x^2(2[y]\{y\})}{2x[y]}$

$= x\{y\} = x\left\{\dfrac{1}{x}\right\}$

$\Psi(x) = \dfrac{\{y\}}{[y] + \{y\}} = \dfrac{\{y\}}{y} = x\{y\} = x\left\{\dfrac{1}{x}\right\}$

Thus, $\Phi(x) = \Psi(x) = x\left\{\dfrac{1}{x}\right\}$.

For $x \in (0,1) \setminus \left\{\dfrac{1}{n} : n \in \mathbb{N}\right\}$, $0 < \left\{\dfrac{1}{x}\right\} < 1 \implies 0 < x\left\{\dfrac{1}{x}\right\} < x$.

By the Squeeze Theorem, as $x \to 0^+$, $u = x\left\{\dfrac{1}{x}\right\} \to 0^+$.

Substituting $u = \Phi(x) = \Psi(x)$ into the limit expression: $L = \lim\limits_{u \to 0^+}\left(\dfrac{2^u + 8^u + 32^u}{3}\right)^{\frac{1}{u}}$

This is an indeterminate form of 1^∞. Taking natural logarithm:

$\ln L = \lim\limits_{u \to 0^+} \dfrac{1}{u}\ln\left(1 + \dfrac{(2^u - 1) + (8^u - 1) + (32^u - 1)}{3}\right)$

$\ln L = \dfrac{1}{3}\left(\lim\limits_{u \to 0^+}\dfrac{2^u - 1}{u} + \lim\limits_{u \to 0^+}\dfrac{8^u - 1}{u} + \lim\limits_{u \to 0^+}\dfrac{32^u - 1}{u}\right)$

$\ln L = \dfrac{\ln 2 + \ln 8 + \ln 32}{3} = \dfrac{\ln(2 \cdot 8 \cdot 32)}{3}$

$= \dfrac{\ln(512)}{3} = \dfrac{9\ln 2}{3} = 3\ln 2 = \ln 8$

$\implies L = 8$

Q3. Let $f : (0, \infty) \to \mathbb{R}$ be a differentiable function defined by

$$f(x) = \int e^x\left(\frac{x^2 - x + 1}{(x^2+1)^{3/2}}\right)dx + \int \frac{x^2 + 1 + (1 - x^2)\ln x}{(x^2+1)^2}dx \text{ such that}$$

$f(1) = \dfrac{e}{\sqrt{2}} + 2$.

Consider the function $g : (0, \infty) \to \mathbb{R}$ defined by

$$g(x) = (x^2+1)f(x) - e^x\sqrt{x^2+1}.$$

If $g(x)$ is strictly increasing at $x = \dfrac{1}{2}$ and $g'\left(\dfrac{1}{2}\right) = k - \ln 2$, then the value of the integer k is

_____.

Ans. 3

Sol. For the first integral, let $u(x) = \dfrac{1}{\sqrt{x^2+1}}$, then $u'(x) = -\dfrac{x}{(x^2+1)^{3/2}}$.

The integrand can be written as $e^x(u(x)+u'(x))$, yielding: $\displaystyle\int e^x\left(\frac{x^2-x+1}{(x^2+1)^{3/2}}\right)dx = \frac{e^x}{\sqrt{x^2+1}}$

For the second integral, let $v(x) = \dfrac{\ln x}{x^2+1}$, then $v'(x) = \dfrac{x^2+1-2x^2\ln x}{x(x^2+1)^2}$.

The integrand matches the form $v(x)+xv'(x)$, yielding: $\displaystyle\int \frac{x^2+1+(1-x^2)\ln x}{(x^2+1)^2}dx = \frac{x\ln x}{x^2+1}$

$f(x) = \dfrac{e^x}{\sqrt{x^2+1}} + \dfrac{x\ln x}{x^2+1} + C$

Using the initial condition $f(1) = \dfrac{e}{\sqrt{2}} + 2$, we substitute $x = 1$ to find $C = 2$.

$g(x) = (x^2+1)\left(\dfrac{e^x}{\sqrt{x^2+1}} + \dfrac{x\ln x}{x^2+1} + 2\right) - e^x\sqrt{x^2+1}$

$= x\ln x + 2x^2 + 2$

$g'(x) = \ln x + 1 + 4x$

$g'\left(\dfrac{1}{2}\right) = \ln\left(\dfrac{1}{2}\right) + 1 + 4\left(\dfrac{1}{2}\right) = 3 - \ln 2$

Since $3 - \ln 2 > 0$, the function $g(x)$ is strictly increasing at $x = \dfrac{1}{2}$.

Comparing $g'\left(\dfrac{1}{2}\right) = 3 - \ln 2$ with $k - \ln 2$ gives $k = 3$.

Q4. Consider two parallel lines $L_1 : 3x - 4y + 7 = 0$ and $L_2 : 3x - 4y - 3 = 0$ in the xy-plane. A circle C is inscribed between L_1 and L_2 such that it is tangent to both lines. Let L_3 be another line parallel to L_1 and distinct from L_2, such that the distance between L_1 and L_3 is equal to the distance between L_1 and L_2. From a variable point P on the line L_3, a pair of tangents PA and PB are drawn to the circle C, where A and B are the points of contact. As P varies along L_3, the chord of contact AB always passes through a fixed point Q. If d_1 and d_2 denote the perpendicular distances of the point Q from the lines L_1 and L_2 respectively, then the value of the ratio $\dfrac{d_2}{d_1}$ is _____.

Ans. 2

Sol. The distance between the parallel lines L_1 and L_2 is given by Distance $= \dfrac{|7-(-3)|}{\sqrt{3^2+(-4)^2}} = \dfrac{10}{5} = 2$

Since the circle C is tangent to both L_1 and L_2, its diameter is 2, yielding a radius $R = 1$.

The centerline (locus of the center) of the circle is the midline of L_1 and L_2, given by $3x - 4y + 2 = 0$.

The line L_3 is parallel to L_1 at a distance of 2 units. Since $L_3 \neq L_2$, its equation must be $3x - 4y + 7 + 10 = 0 \implies 3x - 4y + 17 = 0$

Let us project the geometry onto a coordinate axis perpendicular to the lines, taking the center of the circle as the origin O.

The positions of the lines along this axis are: Center of circle $C = 0$; Line $L_2 = -1$; Line $L_1 = +1$; Line $L_3 = +3$

The variable point P lies on L_3, so the chord of contact AB always passes through the pole of the line L_3 with respect to the circle C.

The distance of L_3 from the center is $d = 3$.

By the properties of poles and polars, the fixed point Q (the pole) lies on the same perpendicular axis at a position given by $OQ = \dfrac{R^2}{d} = \dfrac{1^2}{3} = \dfrac{1}{3}$

Thus, the position of Q on this axis is $+\dfrac{1}{3}$.

$d_1 = \left|1 - \dfrac{1}{3}\right| = \dfrac{2}{3}$; $d_2 = \left|\dfrac{1}{3} - (-1)\right| = \dfrac{4}{3}$

$\dfrac{d_2}{d_1} = \dfrac{4/3}{2/3} = 2$

Q1. Let $f : \mathbb{R} \to \mathbb{R}$ be a function defined by:

$f(x) = \max\left\{x, \min\left\{x^2, 2 - x^2\right\}\right\}$

If N is the number of points in $\mathbb{R}$ where $f(x)$ is not differentiable, then the value of N is

Ans. 3

Sol. Let $g(x) = \min\left\{x^2, 2 - x^2\right\}$.

To find the critical points, set $x^2 = 2 - x^2 \Rightarrow x^2 = 1 \Rightarrow x = \pm 1$.

Thus, $g(x)$ can be written piece-wise as:

$$g(x) = \begin{cases} 2 - x^2, & x < -1 \\ x^2, & -1 \le x \le 1 \\ 2 - x^2, & x > 1 \end{cases}$$

Now, consider $f(x) = \max\{x, g(x)\}$ across different intervals:

1. For $x > 1$: $g(x) = 2 - x^2$.
   Comparing x and $2 - x^2$, we have $x - (2 - x^2) = (x + 2)(x - 1) > 0$, so
   $x > 2 - x^2 \implies f(x) = x$.
2. For $0 \le x \le 1$: $g(x) = x^2$.
   Comparing x and x^2, we have $x \ge x^2 \implies f(x) = x$.
3. For $-1 \le x < 0$: $g(x) = x^2$.
   Since $x^2 \ge 0 > x$, $f(x) = x^2$.
4. For $x < -1$: $g(x) = 2 - x^2$.
   Comparing x and $2 - x^2$, we look at $x - (2 - x^2) = (x + 2)(x - 1)$.
   * If $-2 \le x < -1$, then $(x + 2)(x - 1) \le 0$
   $\Rightarrow 2 - x^2 \ge x \Rightarrow f(x) = 2 - x^2$.
   * If $x < -2$, then $(x + 2)(x - 1) > 0$
   $\Rightarrow x > 2 - x^2 \Rightarrow f(x) = x$.

Combining the regions, the explicit form of $f(x)$ is:

$$f(x) = \begin{cases} x, & x \leq -2 \\ 2 - x^2, & -2 < x \leq -1 \\ x^2, & -1 < x \leq 0 \\ x, & x > 0 \end{cases}$$

Function $f(x)$ is continuous everywhere.

Differentiating $f(x)$ in each interval gives:

$$f'(x) = \begin{cases} 1, & x < -2 \\ -2x, & -2 < x < -1 \\ 2x, & -1 < x < 0 \\ 1, & x > 0 \end{cases}$$

Checking the left-hand derivative (LHD) and right-hand derivative (RHD) at the transition points:

- At $x = -2$: LHD $= 1$, RHD $= 4 \implies$ Non-differentiable.
- At $x = -1$: LHD $= 2$, RHD $= -2 \implies$ Non-differentiable.
- At $x = 0$: LHD $= 0$, RHD $= 1 \implies$ Non-differentiable.

Thus, $f(x)$ is non-differentiable at exactly 3 points ($x = -2, -1, 0$). Hence, $N = 3$.

Q2. Consider the function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = 2x^3 - 6x^2 + 4x$. Let $P(x_0, y_0)$ be the point on the curve $y = f(x)$ where the curve changes its concavity. Let C be another curve defined by $y = \ln(2x - 1)$ for $x > \dfrac{1}{2}$. If θ is the acute angle of intersection between the curve $y = f(x)$ and the curve C at the point P, then the value of $3\tan\theta$ is equal to ____.

Ans. 4

Sol. Given the function:

$f(x) = 2x^3 - 6x^2 + 4x$

Differentiating twice with respect to x:

$f'(x) = 6x^2 - 12x + 4$

$f''(x) = 12x - 12 = 12(x - 1)$

The curve changes its concavity where $f''(x)$ changes its sign, which occurs uniquely at $x = 1$. At $x = 1$, the y-coordinate is $y = f(1) = 2 - 6 + 4 = 0$.

Thus, the point P is $(1, 0)$.

For the second curve $C : y = \ln(2x - 1)$, substituting $x = 1$ yields $y = \ln(1) = 0$, which confirms that $P(1, 0)$ is a point of intersection for both curves.

The slope of the tangent to the first curve at $P(1, 0)$ is:

$m_1 = f'(1) = 6(1)^2 - 12(1) + 4 = -2$

For the second curve C, differentiating with respect to x gives:

$$\frac{dy}{dx} = \frac{2}{2x - 1}$$

The slope of the tangent to C at $P(1, 0)$ is:

$$m_2 = \frac{2}{2(1) - 1} = 2$$

The acute angle θ between the two curves is evaluated as:

$$\tan\theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$
$$= \left| \frac{-2 - 2}{1 + (-2)(2)} \right|$$
$$= \frac{4}{3}$$

Therefore, the required value is:

$$3\tan\theta = 3 \times \frac{4}{3} = 4$$

Q3. Consider the circle $x^2 + y^2 - 4x - 6y - 12 = 0$ and the straight line $3x + 4y = 83$. Let P be a variable point on this straight line. Two tangents PA and PB are drawn from P to the circle, where A and B are the points of tangency. Let C denote the center of the circle. If the minimum area of the quadrilateral $CAPB$ is Δ, then the value of Δ is _____.

Ans. 60

Sol. The given equation of the circle is $x^2 + y^2 - 4x - 6y - 12 = 0$.

The center of the circle is $C(2, 3)$ and its radius R is given by:

$$R = \sqrt{(-2)^2 + (-3)^2 - (-12)} = \sqrt{4 + 9 + 12} = 5$$

The perpendicular distance d from the center $C(2, 3)$ to the straight line $3x + 4y - 83 = 0$ represents the shortest distance from C to the line:

$$d = \frac{|3(2) + 4(3) - 83|}{\sqrt{3^2 + 4^2}}$$
$$= \frac{|6 + 12 - 83|}{5} = \frac{65}{5} = 13$$

Since P lies on the given straight line, the distance $CP \geq 13$.

The quadrilateral $CAPB$ is formed by two congruent right-angled triangles $\triangle CAP$ and $\triangle CBP$, right-angled at A and B respectively.

Therefore, the area of the quadrilateral is:

$$\text{Area} = 2 \times \left(\frac{1}{2} \times CA \times AP \right) = R \times L$$

where L is the length of the tangent AP.

From the right-angled triangle $\triangle CAP$:

$$L = \sqrt{CP^2 - R^2} = \sqrt{CP^2 - 25}$$

Thus, the area of the quadrilateral is expressed as:

$$\text{Area} = 5\sqrt{CP^2 - 25}$$

To minimize the area of the quadrilateral, the distance CP must be minimized.

The minimum value of CP is the perpendicular distance $d = 13$.

Substituting $CP = 13$ into the area expression gives the minimum area Δ:

$\Delta = 5\sqrt{13^2 - 25}$

$= 5\sqrt{169 - 25} = 5 \times 12 = 60$